# Homework Assignment 2, Question 3: Classical and Asymptotic Diffusion against the Exact Time-Dependent Planar Solution

Tal Ramot

August 10, 2026

## 1 The three solutions being compared

Throughout, $\Sigma_t = v = 1$ as specified in the assignment, the source is the plane pulse $Q(x,t) = \delta(x)\delta(t)$, and the figures are generated by `src/homework3`.

The **exact** solution is the planar-source scalar flux derived in Question 2 from the Paasschens point-source solution, carried to general $c$ by the scaling identity of Question 1:

$$\phi(x,t;1) = \frac{e^{-t}}{2t}\left[1 + \sqrt{\frac{3}{\pi}}\int_0^{w_0} \sqrt{w}\, G(w)\, dw\right]\Theta(t-|x|), \qquad w_0 = t\left(1-\frac{x^2}{t^2}\right)^{3/4} \tag{1}$$

$$\phi(x,t;c) = c\, e^{-(1-c)t}\, \phi(cx, ct; 1) \tag{2}$$

The two **diffusion** solutions are the Green's functions of

$$\frac{\partial\phi}{\partial t} - D\frac{\partial^2\phi}{\partial x^2} + (1-c)\Sigma_t\phi = \delta(x)\delta(t) \qquad \Longrightarrow \qquad \phi_{\text{diff}}(x,t;c) = \frac{e^{-(1-c)t}}{\sqrt{4\pi Dt}}\, e^{-x^2/(4Dt)} \tag{3}$$

differing only through $D = D_0(c)/\Sigma_t$:

$$D_0 = \tfrac{1}{3} \quad \text{(classical)}, \qquad D_0(c) = (1-c)\nu_0^2 \quad \text{(asymptotic)}, \tag{4}$$

where $\nu_0$ is the discrete transport eigenvalue, $c\nu_0 \operatorname{arctanh}(1/\nu_0) = 1$. For $c > 1$ the eigenvalue is imaginary, $\nu_0 = i|\nu_0|$, and $(1-c)\nu_0^2 = (c-1)|\nu_0|^2$ remains positive; both branches tend to $\frac{1}{3}$ as $c \to 1$, so the two approximations coincide there. Integrating (3) over all $t$ recovers the familiar steady result $\phi(x) = \frac{\sqrt{3}}{2\sqrt{1-c}} e^{-\sqrt{3(1-c)}\Sigma_t|x|}$, which fixes the normalisation of (3).

| $c$ | 0.6 | 0.8 | 1.0 | 1.2 | 1.5 |
|---|---|---|---|---|---|
| $D_0(c)$ | 0.48588 | 0.39629 | 0.33333 | 0.28717 | 0.23745 |

Table 1: The asymptotic diffusion coefficient; the classical value is 1/3 for every $c$.

## 2 Question 3(a) and 3(b): the comparison

Each figure fixes $c$ and shows $t = 1, 2, 3, 4, 7, 15$. The vertical line marks the causal front $|x| = vt$, beyond which the exact solution is identically zero. Only $x \geq 0$ is plotted; all three solutions are even in $x$.

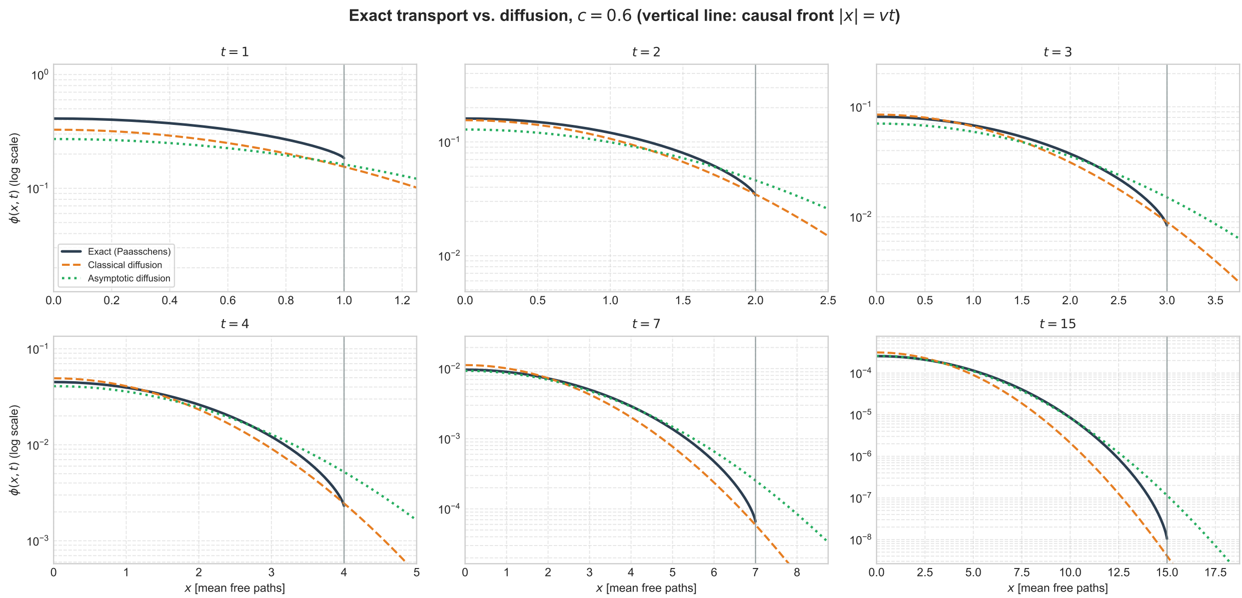


Figure 1: $c = 0.6$.

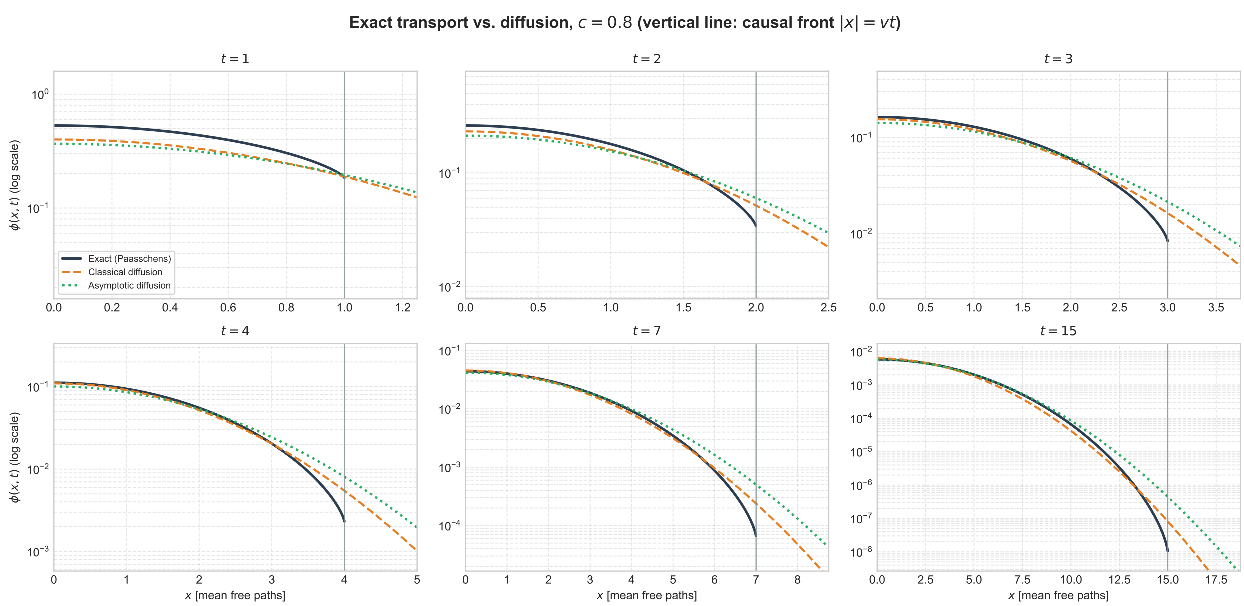


Figure 2: $c = 0.8$.

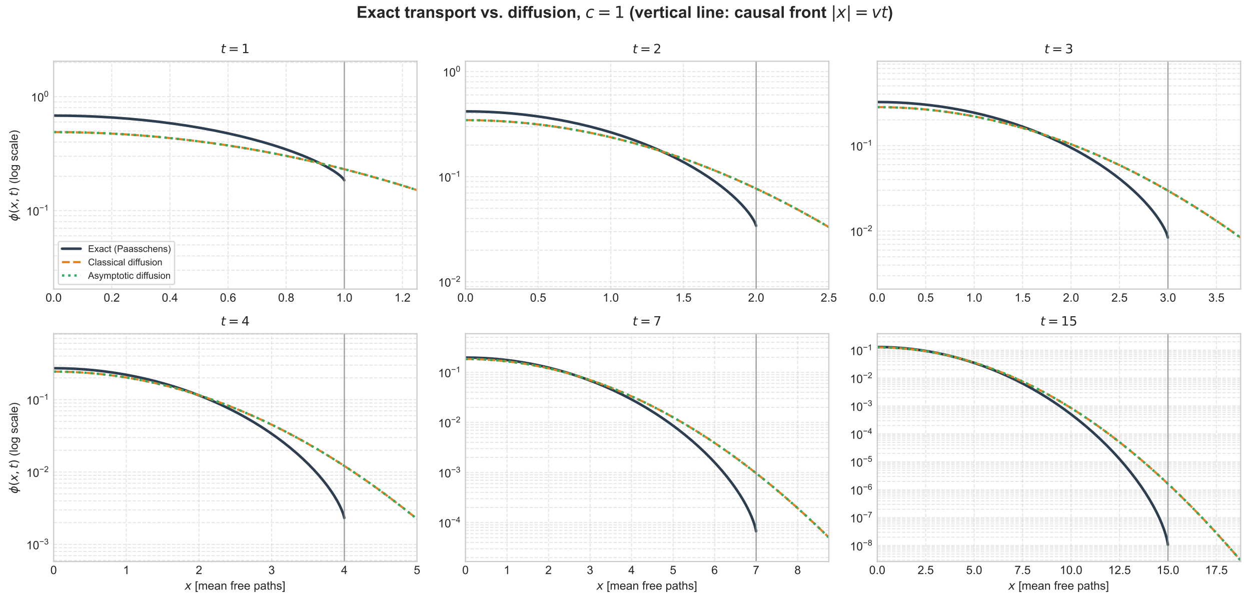


Figure 3: $c = 1$. The two diffusion coefficients coincide at $D_0 = 1/3$, so the classical and asymptotic curves overlie each other.

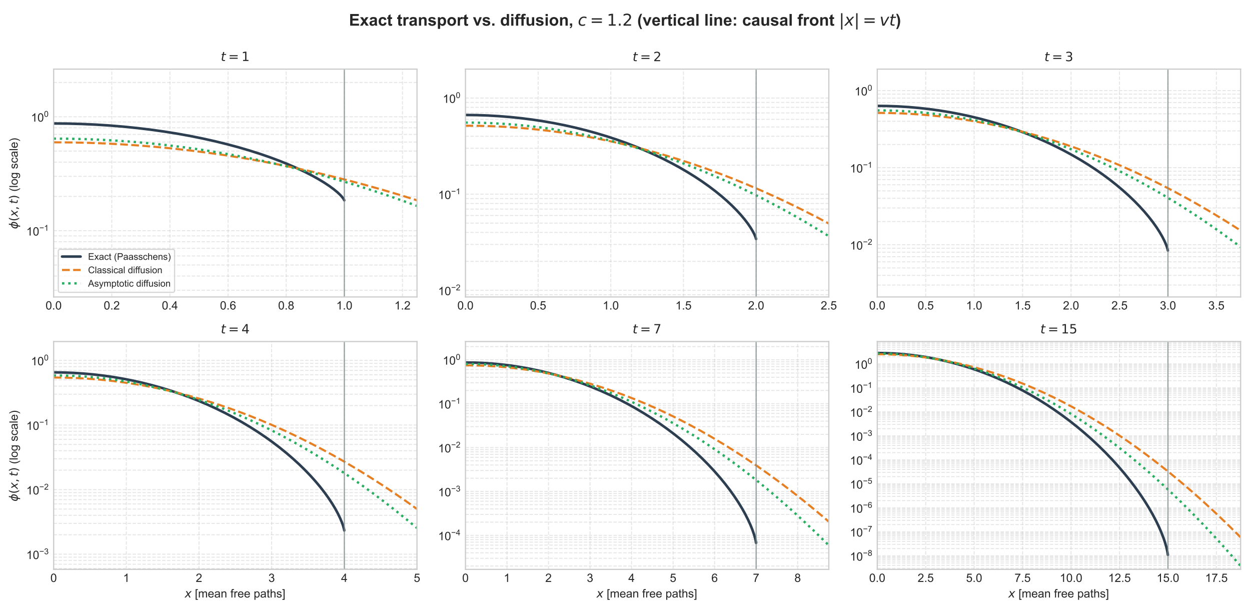


Figure 4: $c = 1.2$.

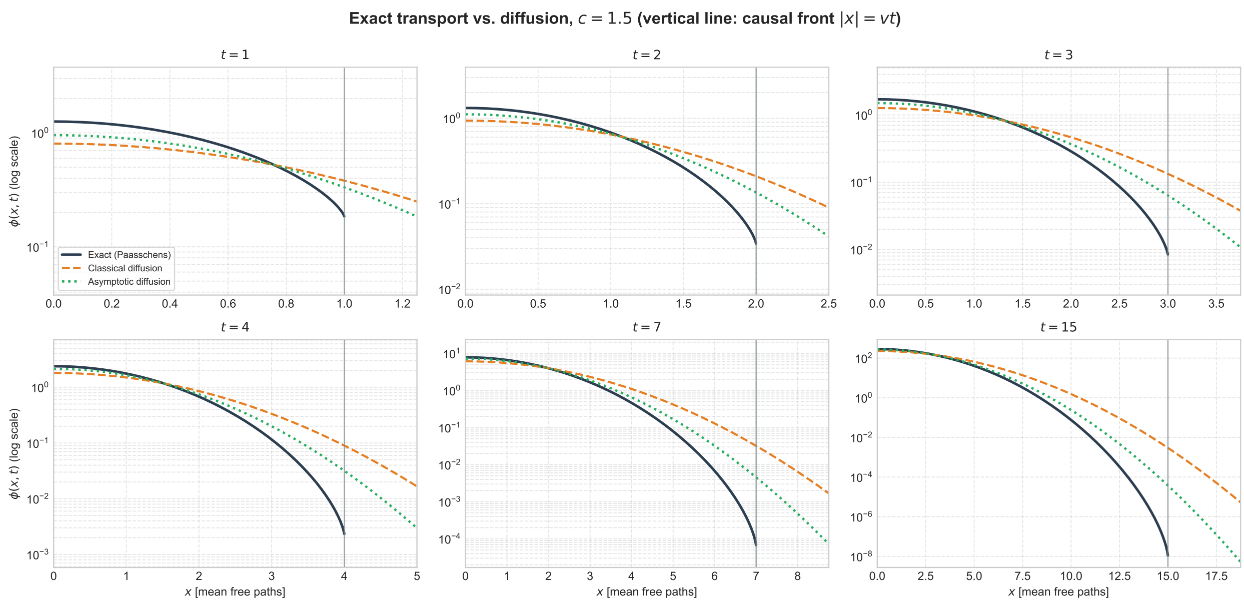
Exact transport vs. diffusion, $c = 1.5$ (vertical line: causal front $|x| = vt$)
$t = 1$
$t = 2$
$t = 3$
$t = 4$
$t = 7$
$t = 15$
$\phi(x, t)$ (log scale)
$x$ [mean free paths]
Exact (Paasschens)
Classical diffusion
Asymptotic diffusion

Figure 5: $c = 1.5$.